

Global Income Inequality Analysis



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Income inequility distributed across countries

Most Countries Cluster in Mid-Range

- Majority of values lie between **30 and 50**
Indicates that most countries have **moderate inequality**

2. Peak Around 30–40

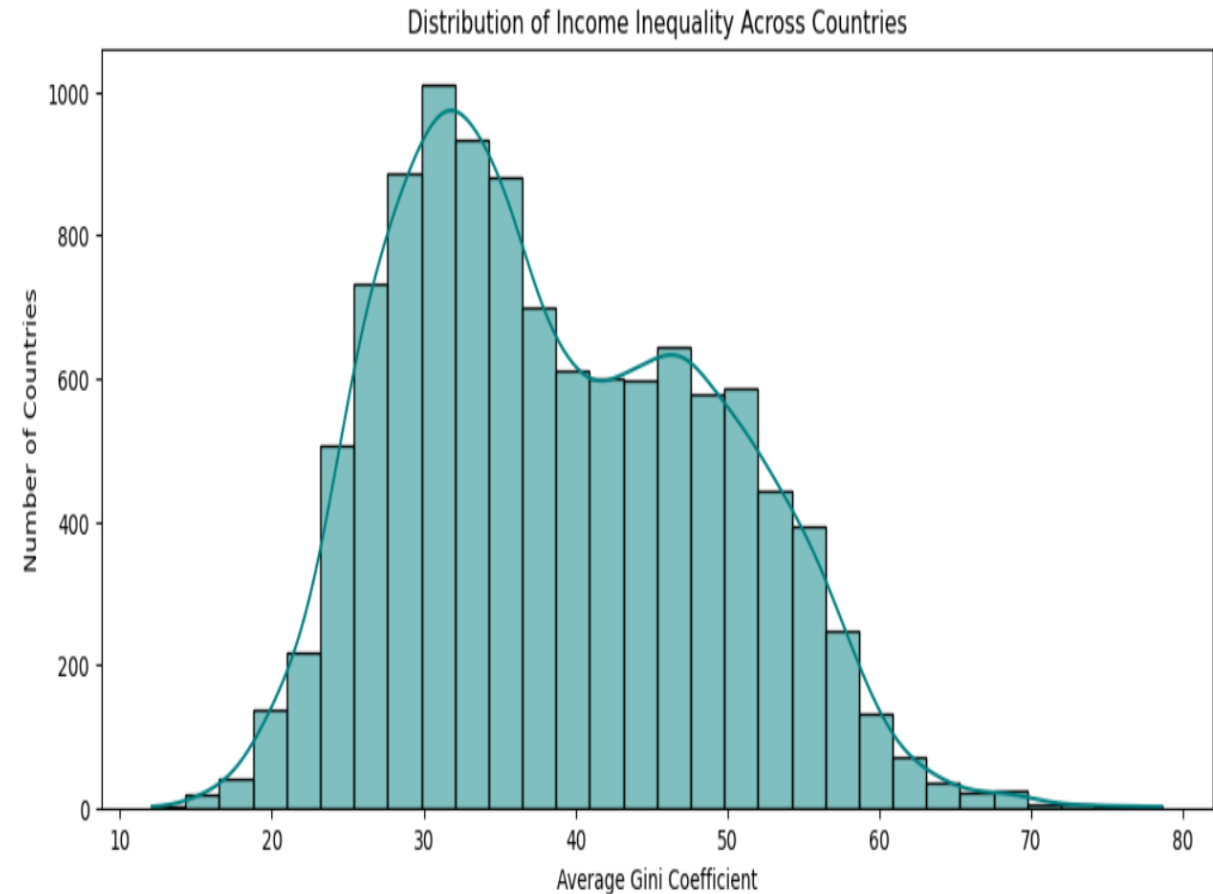
- Highest frequency is usually around **30–40 range**
This is the **most common level of inequality globally**

3. Fewer Extremely Low or High Values

- Very low (<25) and very high (>60) Gini values are **rare**
Extreme equality or inequality is uncommon

4. Slight Right Skew (Positive Skew)

- Tail extends toward higher values (50–80)
Some countries have **very high inequality**, but fewer in number



Variation of Income inequality across regions

Africa

- **Highest median (~45)**
- Very wide range (up to ~80)
- Many extreme values
- **Most unequal and most variable region**

Americas

- High median (~47–48)
- Moderate spread
- **Consistently high inequality**

Asia

- Median around 35
- Moderate spread with some high outliers

Mixed inequality levels

Europe

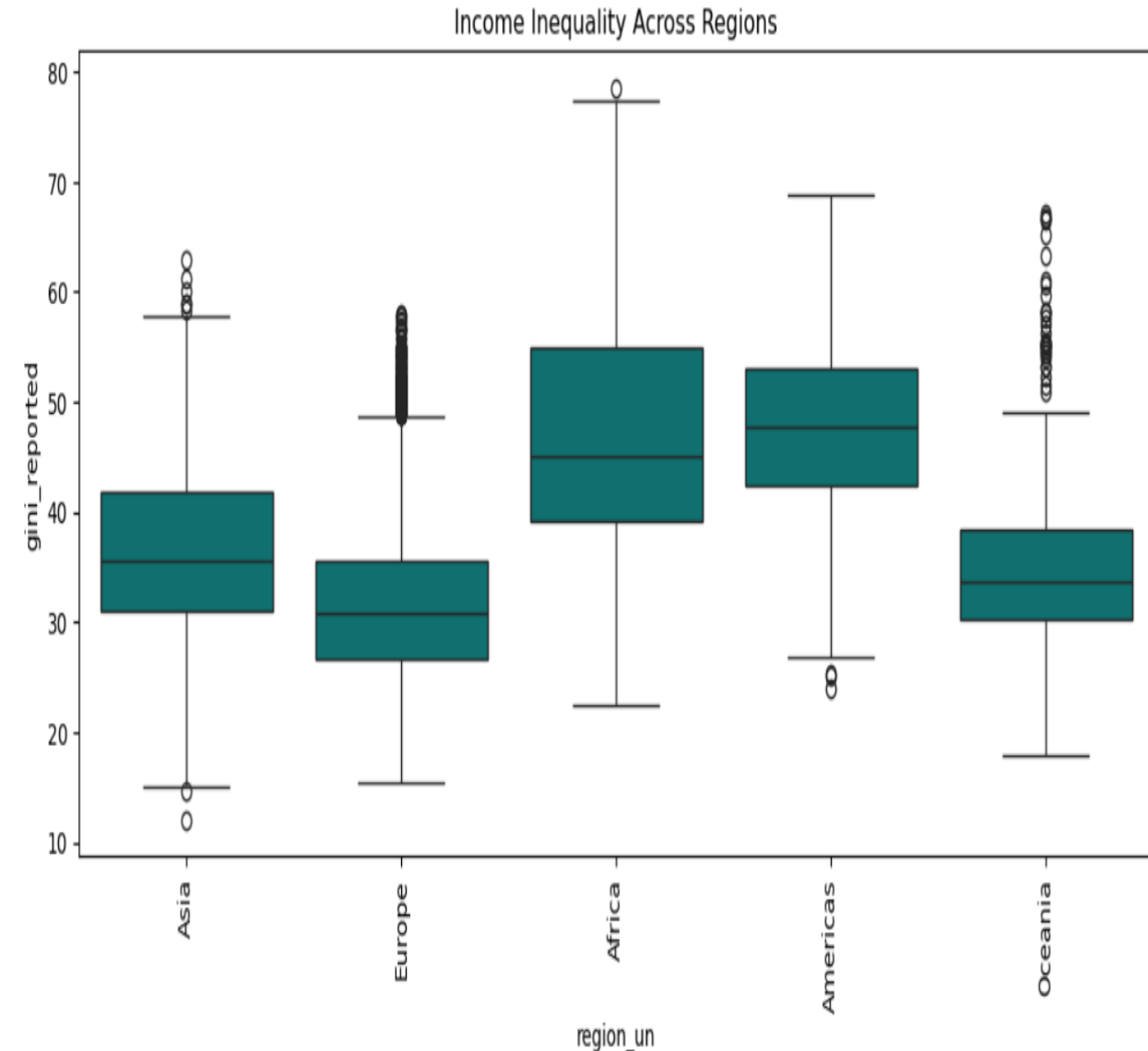
- **Lowest median (~30)**
- Narrow spread

Most equal and consistent region

Oceania

- Median around 33–35
- Some high outliers

Moderate inequality with variation



Are there noticeable regional patterns in Income distribution

Africa

- Highest median (~45–50)
- Wide spread (roughly 30 to 70+)

Most unequal region overall, with large variation

Americas

- High median (~47–50)
- Distribution is concentrated at higher values

Consistently high inequality

Asia

- Moderate median (~33–35)
- Wide distribution

Mix of both **low and high inequality countries**

Europe

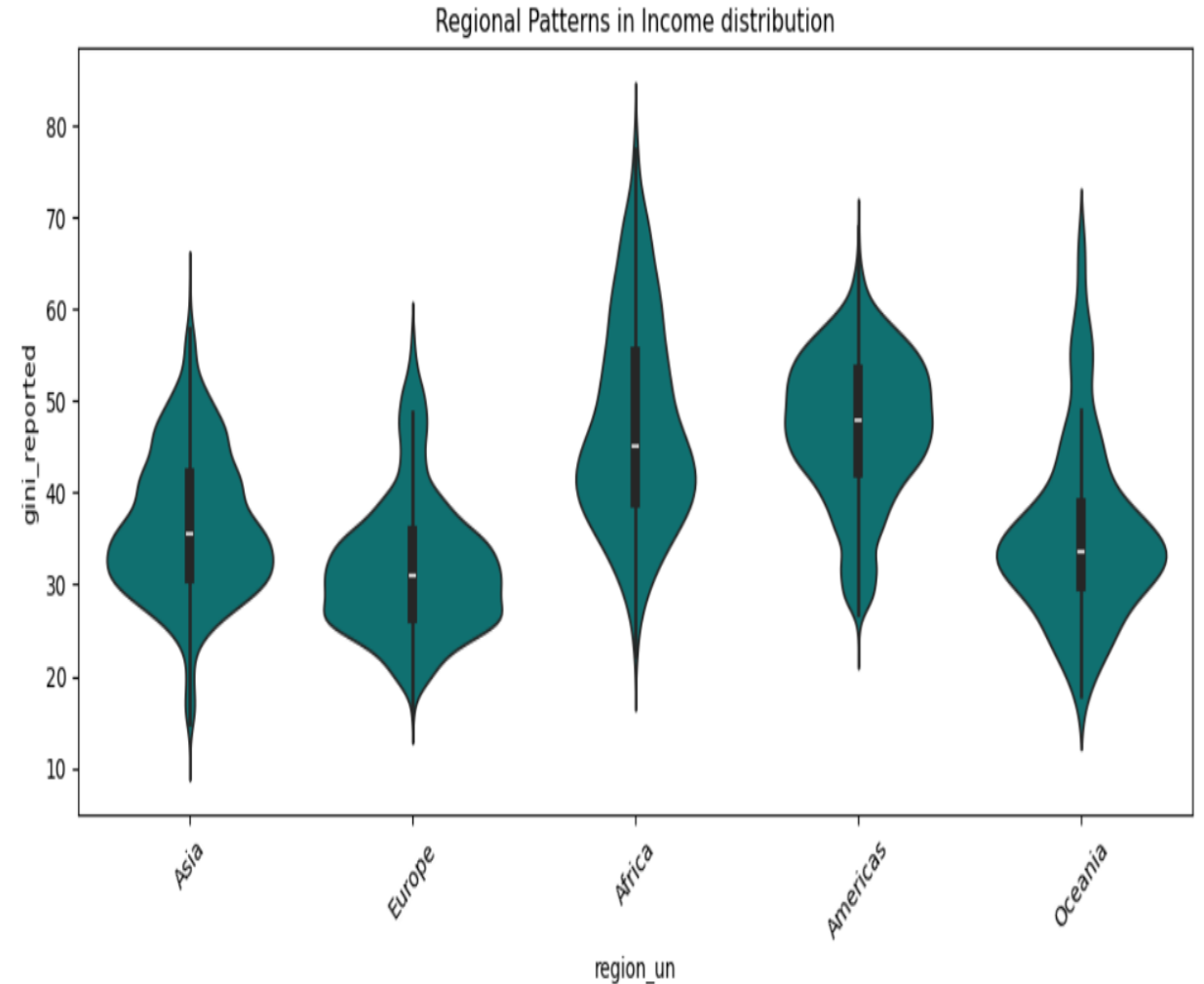
- Lowest median (~28–32)
- Narrow spread

Most equal region, with consistent low inequality

Oceania

- Moderate median (~32–35)
- Some variation

Generally **moderate inequality**



Relationship between GDP per capita and Income inequality

--The relationship between **GDP per capita** and **income inequality (Gini coefficient)** is **not perfectly linear**, but a **general pattern is visible**.

1. Negative Trend (Overall)

- As **GDP per capita** increases, the **Gini coefficient** tends to **decrease**

- This suggests:

Wealthier countries generally have **lower inequality**

2. High Variation at Low Income Levels

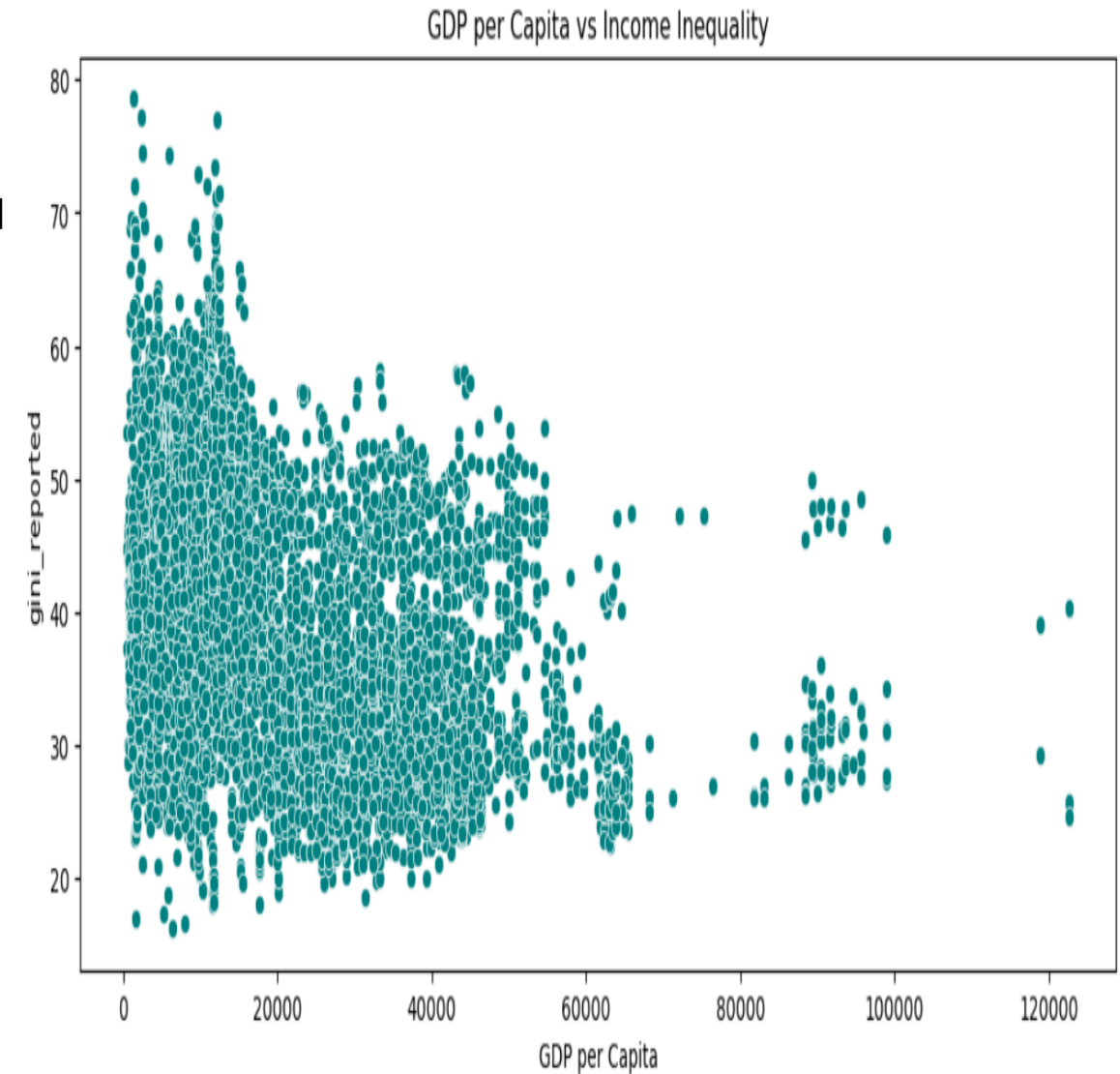
- At **low GDP levels**, inequality ranges widely (~20 to 80)
- Some poor countries are relatively equal, while others
- are extremely unequal

3. More Stability at High Income Levels

- At **high GDP levels**, most values cluster between **25–40**
- Inequality becomes more **consistent and moderate**

4. Presence of Exceptions

- Some high-income countries still show **moderate inequality (~45–50)**
- So, wealth does **not guarantee equality**



Inequality variation between income groups



- **Low income** ≈ 44–45 (highest inequality)
- **Upper middle income** ≈ 44–45 (almost same as low income)
- **Lower middle income** ≈ 41–42
- **High income** ≈ 34–35 (lowest inequality)

Interpretation:

There is a **clear downward trend**:

- As income level increases → **average inequality decreases**

• **High-income countries** have **significantly lower inequality**

However: -- **Low income and upper middle income** are very similar

-- So the pattern is **not perfectly consistent** across all groups

Do wealthier nations consistently maintain lower inequality levels?

No—wealthier nations **generally** have lower inequality, but they do **not consistently maintain** it.

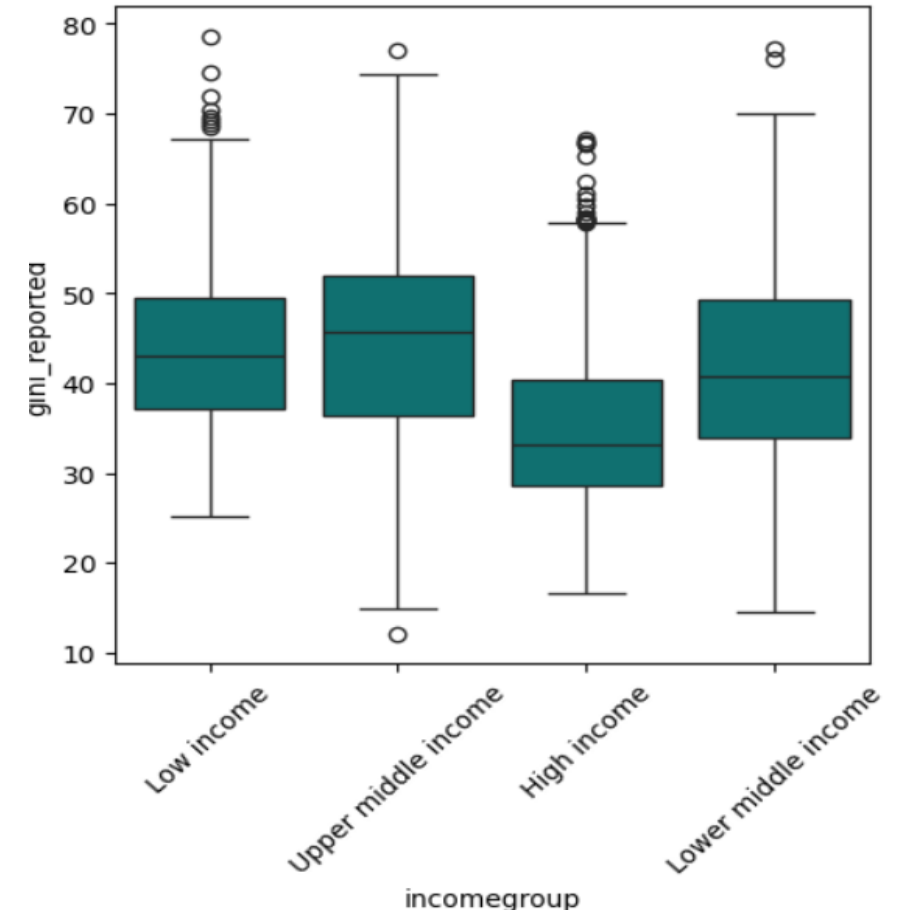
• **High-income countries** have a **lower median Gini** (less inequality), showing a general trend toward more equal income distribution.

However, there are **outliers and overlaps**:

- Some high-income countries still show **moderate to high inequality**.
- Some middle- and low-income countries have **similar or even lower inequality levels** than certain wealthy nations.

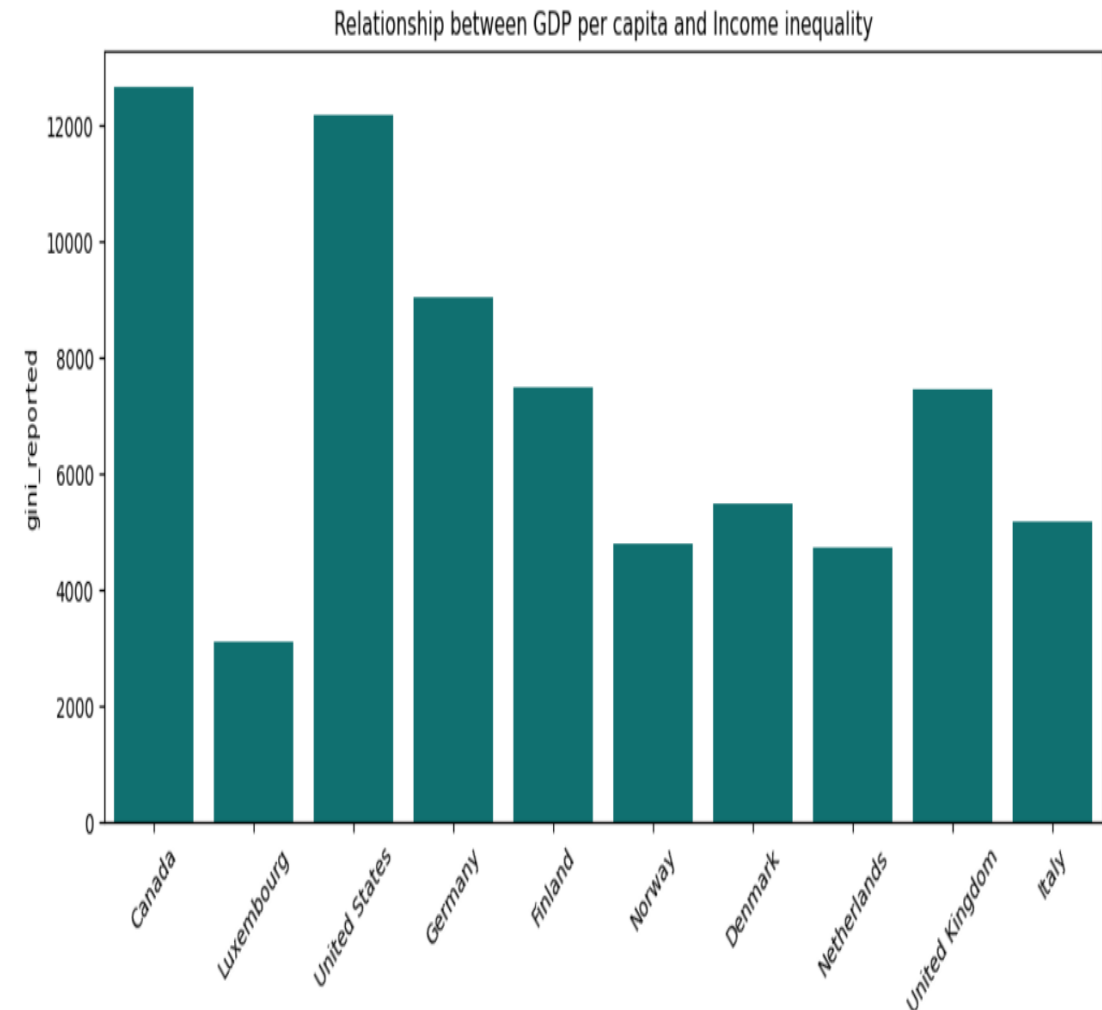
Key Insight:

- The distributions **overlap across income groups**, meaning income level alone does not fully determine inequality.
- Other factors (government policies, taxation, social welfare, etc.) also play a major role.

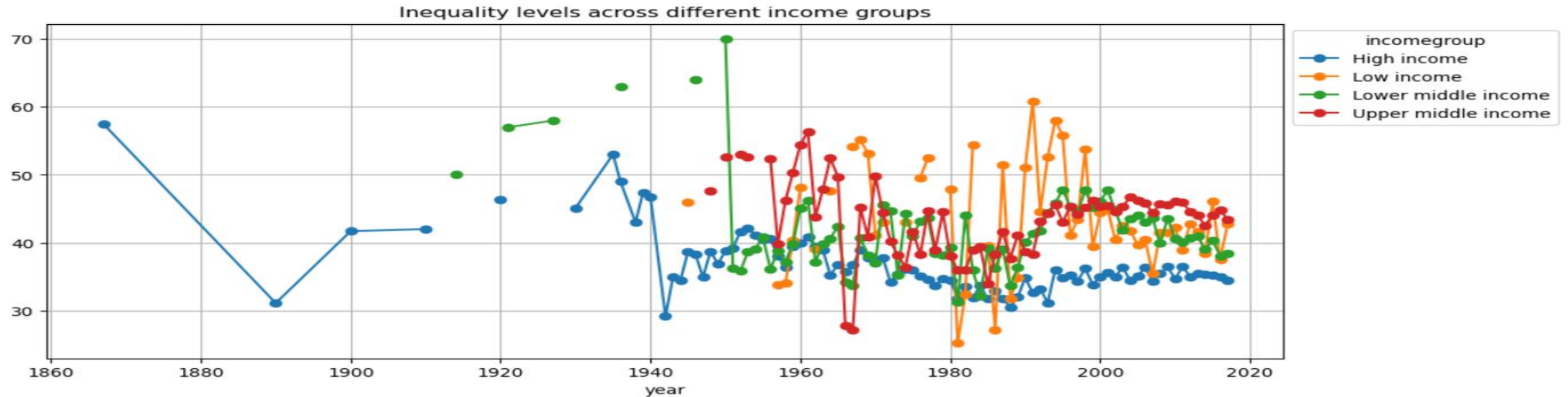


Is there any relationship between GDP per capita and Income inequality?

- **GDP per Capita vs Inequality Insights**
- **Overall Observation**
- The graph compares **GDP per capita with income inequality**
- No strict linear relationship → higher GDP doesn't always mean lower inequality
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- **Key Country Insights**
- **United States & Canada**
 - High GDP + **high inequality**
 - Wealth exists but is **unevenly distributed**
- **Luxembourg**
 - Very high GDP + **low inequality**
 - Shows **strong wealth distribution** despite high income
- **Germany & United Kingdom**
 - Moderate to high GDP + **moderate inequality**
 - Balanced but still some income gaps
- **Nordic Countries (Norway, Denmark, Finland)**
 - High GDP + **low inequality**
 - Strong **welfare and redistribution** systems
- **Netherlands & Italy**
 - Moderate GDP + **moderate inequality**
 - Stable but not as equal as Nordic countries

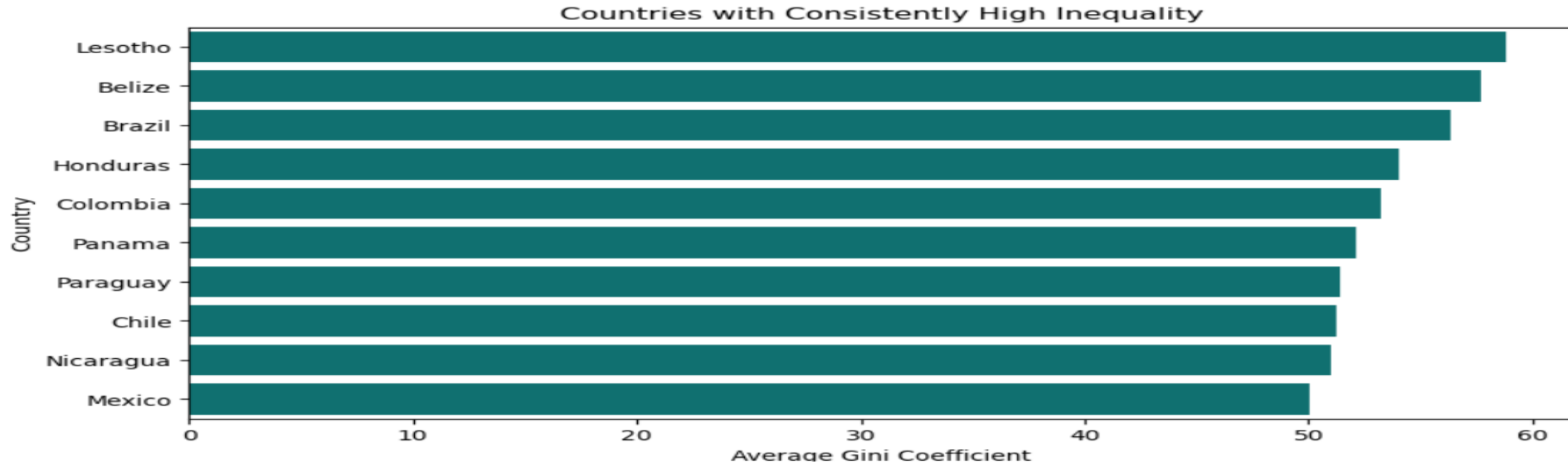


How stable are inequality levels across different income groups



- **Low-income countries** show the **highest inequality** (often 45–60+)
- **Lower middle-income countries** also have **high but slightly lower inequality**
- **Upper middle-income countries** show **moderate inequality** (~40–50)
- **High-income countries** have the **lowest and most stable inequality** (~30–40)
- **Overall:**
As income level increases, **inequality generally decreases and becomes more stable.**

Which countries consistently show high inequality levels?



- Countries like **Lesotho, Belize, Brazil** show **consistently very high inequality (above 55)**
- Most countries in the list have Gini values **above 50**, indicating persistent inequality
- Many are from **Latin America and Africa**, suggesting regional inequality patterns

Overall:

These countries face **long-term structural inequality**, meaning the income gap remains high over time and is not just temporary.

Which Countries show the highest and lowest inequality levels

Top 10 Highest Inequality

• Countries like **South Africa, Namibia, Lesotho** have **very high Gini values (~55–63)**

• Insight:

- Indicates **extreme income gap**
- Wealth is concentrated among a small population

Top 10 Lowest Inequality

• Countries like **Czechoslovakia, Slovakia, Andorra** have **low Gini (~20–28)**

• Insight:

- Shows **more equal income distribution**
- Strong welfare systems or balanced economies

Overall Comparison

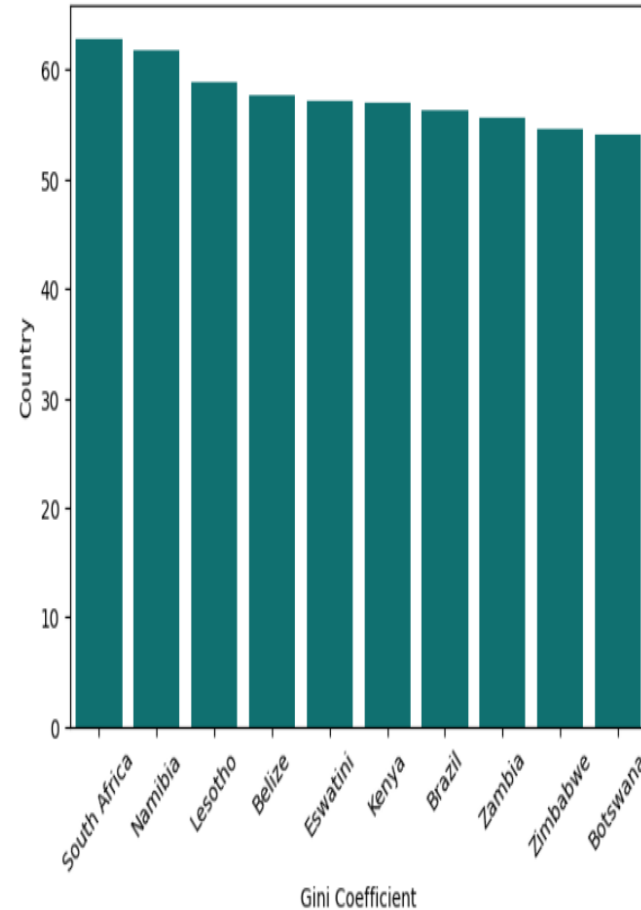
• High inequality countries are mostly **developing regions**

• Low inequality countries are mostly **European nations**

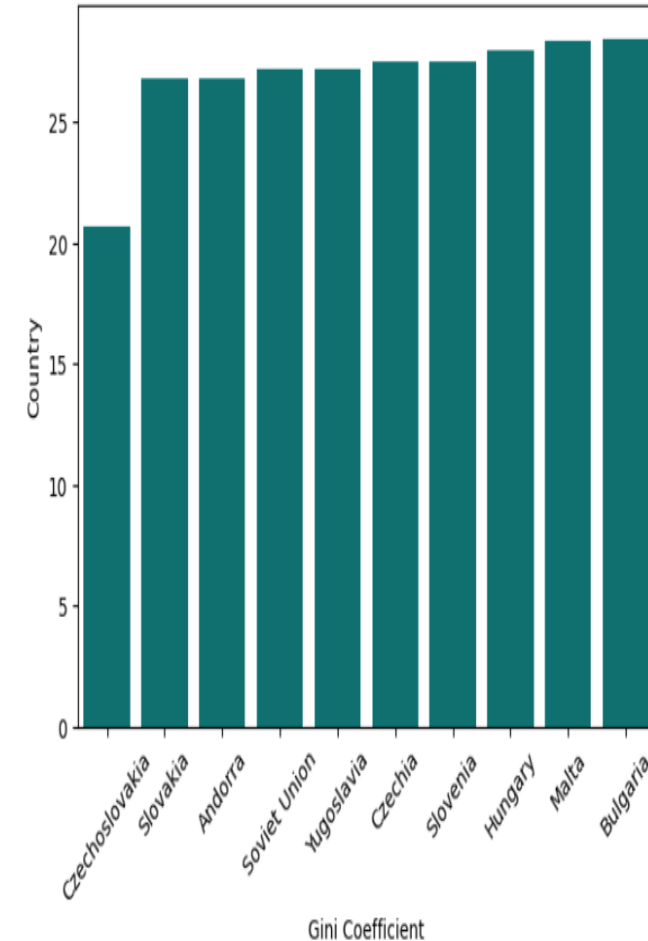
• Final Insight:

- **Economic development and policies play a major role in income distribution**

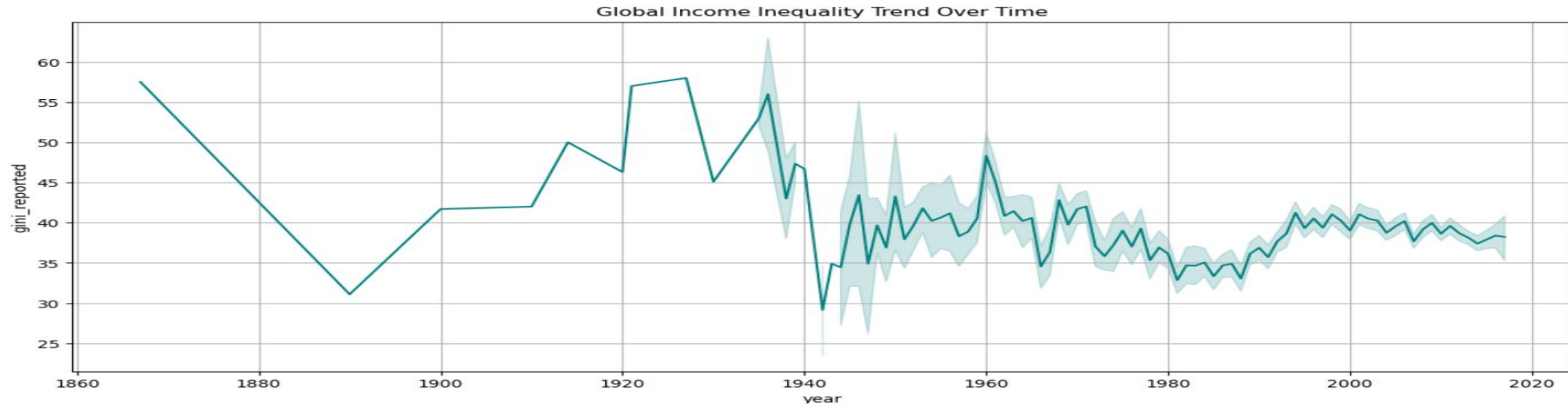
Top 10 Countries with Highest Inequality



Top 10 Countries with Lowest Inequality



How has income inequality changed over time globally



- Income inequality **decreased from 1870–1900**, showing improved distribution.
- It **rose again till the 1920s**, indicating wealth concentration.
- The **1930s–1940s show high fluctuations**, likely due to economic crises.
- From **1950–1980**, inequality **stabilized and slightly declined**.
- After **1990**, it **increased moderately** due to globalization/technology.
- In **recent years**, it **remains stable around moderate levels (~38–40)**.

Overall: Inequality follows a **cyclical pattern** with high instability during crises and relative stability in modern times.

Is there any relationship between a country's population size and inequality level

- **Low population countries** show **very high variation in inequality** (Gini ~15 to 80)
→ meaning small countries can be either very equal or very unequal.
- **Medium population countries** (around 0.3–0.7B) mostly cluster around **30–45 Gini**
→ indicating **moderate and more stable inequality**.
- **High population countries** (above 1B) mostly have **lower to moderate inequality (20–40)**
→ suggests large populations tend to have **more balanced distribution**.
- There is **no strong linear relationship** between population and inequality
→ population size alone does **not determine inequality**.

Overall:

Inequality varies widely in smaller countries, but becomes **more stable and moderate as population increases**, with no direct correlation.

